

Trojan & Remote Access Tool (RAT) Analysis

Controlled Lab Exercise Report

Exercise Focus:	njRAT v0.8d — Remote Administration Tool / Backdoor
Lab Environment:	Isolated CEH training range (domain: CEH.com)
Attacker Host:	10.10.1.11
Target Host:	SERVER2022 (10.10.1.22) — Windows Server 2022 Standard SP0 x64
Exercise Date:	August 2, 2026
Report Classification:	Internal / Training Use Only

This document summarizes a hands-on exercise conducted in an isolated, purpose-built lab environment used for Certified Ethical Hacker (CEH)-style training. All activity was performed against dedicated lab systems with no connection to production or third-party networks. The exercise demonstrates how a Remote Access Trojan (RAT) operates end-to-end and how the resulting artifact can be identified, analyzed, and detected by security tooling.

1. Executive Summary

The exercise walked through the full lifecycle of a Remote Access Trojan (RAT) attack chain using njRAT v0.8d in an isolated lab: building a client/server payload, achieving a simulated compromise of a Windows Server 2022 lab host, exercising post-compromise capabilities (process listing, file browser, registry viewer, and remote chat), and then pivoting to the defensive side of the exercise — submitting the resulting binary to multi-engine antivirus and sandbox analysis services, performing static disassembly/entropy review, and reviewing live network connections on the victim host with TCPView.

Findings confirm that the generated payload is reliably flagged as malicious by the overwhelming majority of antivirus engines and sandbox platforms, and that its network behavior (an outbound TCP session to the attacker's listener) is clearly observable with standard host-based monitoring tools. This aligns with the exercise's learning objective: understanding attacker tradecraft in order to improve detection, monitoring, and user-awareness controls.

2. Exercise Objectives

- Understand how a RAT builder configures a payload (callback host/port, victim naming, persistence options).
- Observe the attacker console view of an active/online victim session.
- Review common post-exploitation capabilities exposed by the tool (process management, file system access, registry access, remote messaging).
- Demonstrate why user awareness matters — a victim-side prompt illustrates how social engineering can lead to inadvertent data disclosure.
- Analyze the resulting payload using public multi-engine AV scanning and sandbox detonation services.
- Perform static analysis (disassembly and entropy/packing check) on a related sample.
- Identify the malware's network fingerprint using a live connection-monitoring utility.

3. Walkthrough & Observations

3.1 Payload Builder Configuration

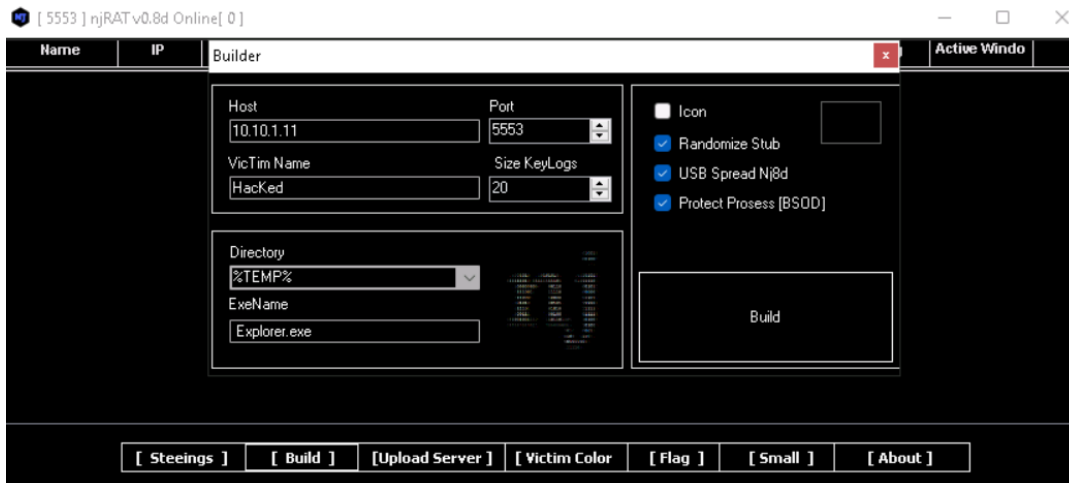


Figure 1 — njRAT v0.8d Builder configuration screen.

- The builder was configured with a callback host of 10.10.1.11 on port 5553, pointing the payload back to the attacker's listener.
- The victim identifier field ('VicTim Name') was set to a lab-only label used to tag the session in the console.
- Optional builder toggles observed: stub randomization, USB propagation, and a process-protection flag intended to resist termination — all disabled/enabled per lab instructions and not used outside the isolated range.

3.2 Establishing the Lab Session



Figure 2 — Attacker console showing the lab target as 'Online' with a live process list.

- Once executed on the lab target, the session registered in the attacker console with host metadata: hostname SERVER2022, user Administrator, OS Windows Server 2022 Standard SP0 x64, source country tagged USA (lab default).
- A remote process manager window lists running processes and PIDs on the target, illustrating the visibility a RAT operator gains into the compromised host.

3.3 Registry Access

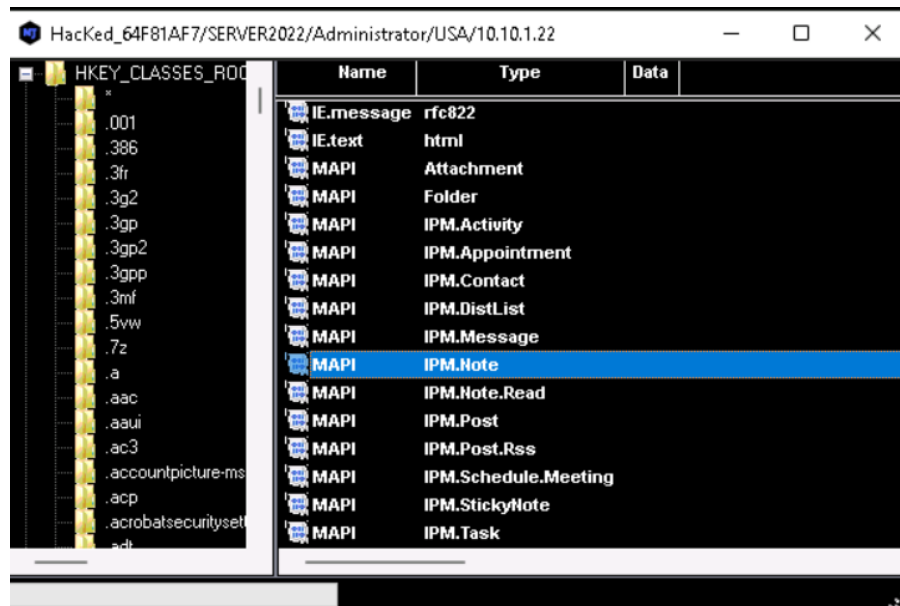


Figure 3 — Remote registry viewer browsing HKEY_CLASSES_ROOT on the lab target.

- The tool exposes a remote registry browser, here showing file-association / MAPI-related keys under HKEY_CLASSES_ROOT.
- This capability highlights the level of system access a successful RAT infection can provide, well beyond simple remote-command execution.

3.4 Target Network Configuration

```

Windows IP Configuration

Host Name . . . . . : Server2022
Primary Dns Suffix . . . . . : CEH.com
Node Type . . . . . : Hybrid
IP Routing Enabled . . . . . : No
WINS Proxy Enabled . . . . . : No
DNS Suffix Search List . . . . . : CEH.com

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Hyper-V Network Adapter
Physical Address. . . . . : 00-15-5D-01-80-02
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80:9d68:1d1a:92eb:e27e%3(Preferred)
IPv4 Address. . . . . : 10.10.1.22(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.10.1.2
DHCPv6 IAID . . . . . : 100668765
DHCPv6 Client DUID . . . . . : 00-01-00-01-29-8D-AD-F9-00-15-5D-01-80-02
DNS Servers . . . . . : 127.0.0.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Hyper-V Network Adapter #2
Physical Address. . . . . : 02-15-5D-29-88-89
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

```

Figure 4 — 'ipconfig /all' output captured from the lab target.

- Confirms the lab target's identity: hostname Server2022, DNS suffix CEH.com (consistent with a dedicated training domain), and IPv4 address 10.10.1.22 on the lab's isolated Hyper-V virtual network.

3.5 Remote File Browser

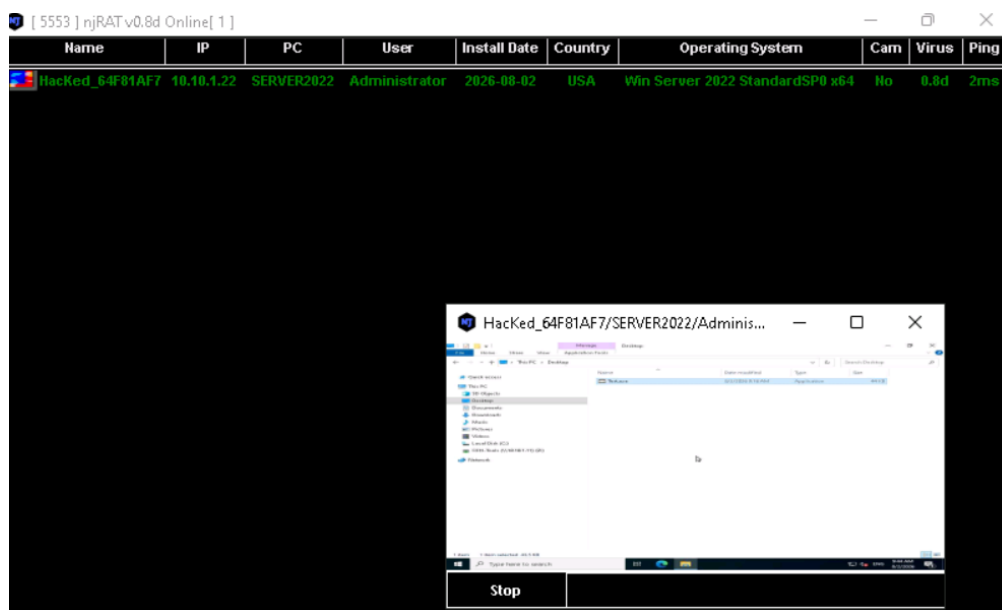


Figure 5 — File-manager view of the target's Desktop folder inside the attacker console.

- The built-in file manager allows the operator to browse the victim's file system, here showing the Desktop of the compromised lab account.

3.6 Simulated Social Engineering / Data Exposure

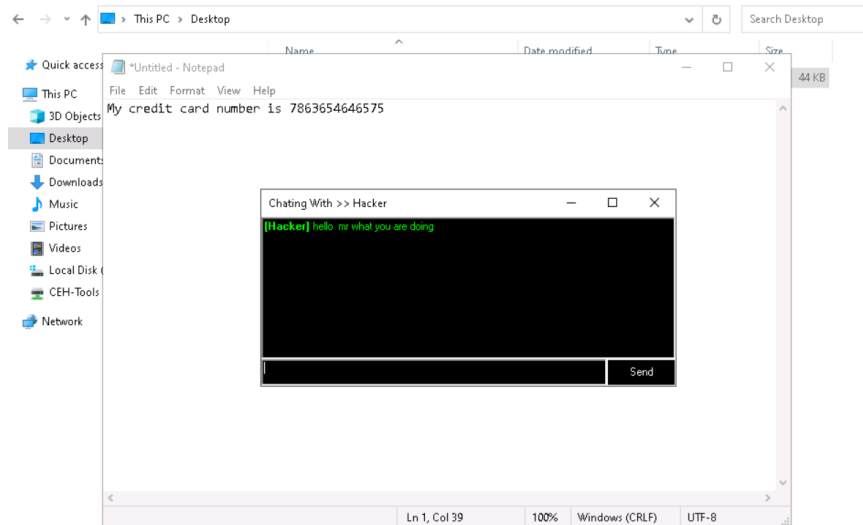


Figure 6 — Lab victim machine with a remote chat prompt and a Notepad window containing a fabricated payment-card number.

- The tool's built-in chat feature was used to message the (simulated) victim, illustrating how an attacker can directly interact with a compromised user for social-engineering purposes.
- A test/dummy card number was typed into Notepad purely to demonstrate how sensitive data left in plaintext on a compromised endpoint (or a follow-up keylogger capture) could be exposed to an attacker. No real financial data was used.

4. Post-Exercise Malware Analysis

Following the attacker-side walkthrough, the exercise shifted to the defender/analyst perspective, submitting the generated artifact to public detection and analysis services and performing static review to reinforce detection engineering skills.

3.7 Multi-Engine Antivirus Detection

Anti-Virus Scan Results for [OPSWAT Metadefender](#) (23/27) ×
Last update: 2026-07-31 14:38:46 (UTC)

Scrutiny	✗ Malware	Vir.IT eXplorer	✗ Backdoor.Win32.Generic.ANLY
K7	✗ Trojan (0055e3df1)	AhnLab	✗ Win-Trojan/IQ.B
Aurora	✗ Malware_-10	CMC	✗ Backdoor_Win32_Tiny_B
Xcitium	✗ Backdoor.Win32.Tiny.B	RocketCyber	✓
ClamAV	✗ Win.Trojan.Tiny-111	Huorong	✗ Trojan/Generic!C3F0738EBDF0F848
Bitdefender	✗ Gen.Variant.Fragtor.113355	Trellix	✗ BackDoor-IQ.b
Gridinsoft	✓	Avira	✗ TR/W32.Tiny.DU
Filseclab	✗ Tini	Zillya!	✗ Backdoor.TinyWin32.1
Sophos	✗ Troj/Cmdoor-A	VirusBlokAda	✓

Close

Figure 7 — Multi-engine AV scan summary (OPSWAT Metadefender), 23 of 27 engines flagging the sample.

- 23 of 27 scan engines flagged the sample as malicious, with detection labels clustering around generic backdoor/trojan signatures (e.g., 'Backdoor.Win32.Generic', 'Backdoor.Win32.Tiny', 'TR/W32.Tiny.DU').
- The consistent 'Tiny'/'Backdoor' family naming across multiple vendors indicates the sample is a well-known, previously-catalogued RAT stub rather than a novel threat.

3.8 Sandbox Detonation Report

HYBRID ANALYSIS ☰

Analysis Overview

[Request Report Deletion](#) [Show Sample Content](#)

Submission	tini.exe	malicious Threat Score: 100/100 AV Detection: 93% Labeled As: Backdoor.Generic #virus #dog #hashtag #hacking #comment #please #file #virus_analysis View all tags (48) Post Link Email 👍 - 👎 0 Community Score 0
name:		
Size:	3KiB	
Type:	peexe executable ⓘ	
Mime:	application/x-dosexec	
SHA256:	9654bb748199882b0fb29b1fa597c0cfe3b9d610adf4188a0b440f9faf5ee527 ⓘ	
Submitted	2017-08-10 07:49:29 (UTC)	
At:		
Last Anti-Virus Scan:	2026-07-31 14:38:46 (UTC)	
Last Sandbox Report:	2025-05-10 21:23:35 (UTC)	

Anti-Virus Results

Updated 2 days ago - [Click to Refresh](#)

Figure 8 — Hybrid Analysis sandbox report for the sample (tini.exe).

- The sandbox assigned a Threat Score of 100/100 with an overall AV detection rate of 93%, labeling the sample 'Backdoor.Generic'.
- File metadata: a 3 KiB Windows PE executable (application/x-dosexec), confirming the payload is a minimal 'stub' typical of RAT droppers designed to fetch or execute further functionality post-infection.

3.9 Static Disassembly Review

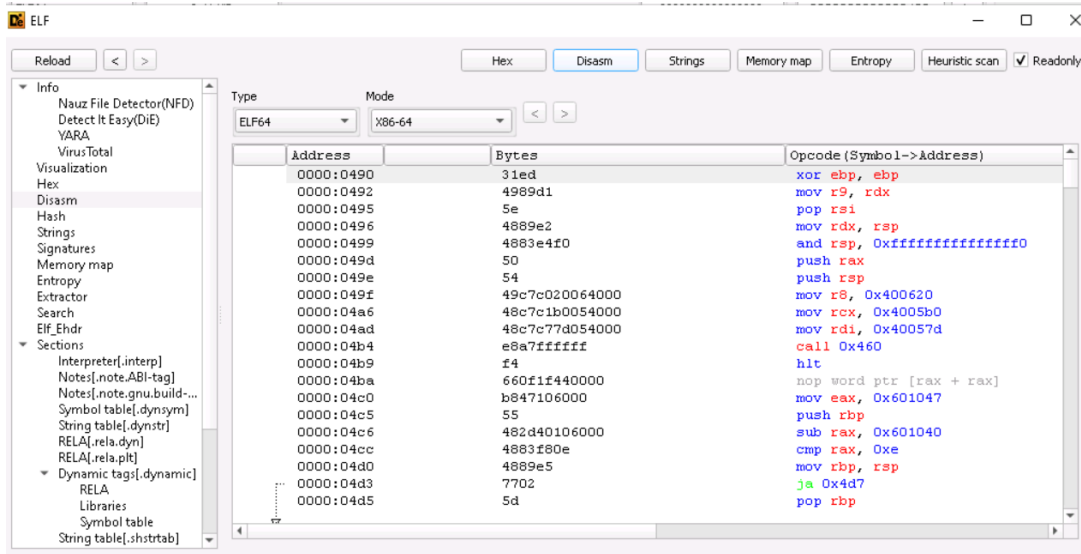


Figure 9 — Disassembly view (Detect It Easy) of the sample's ELF/PE entry point.

- Static analysis tooling was used to inspect the binary's entry point and early control flow (stack setup, calls into the runtime) as part of standard triage prior to deeper reverse engineering.

3.10 Packing / Entropy Analysis

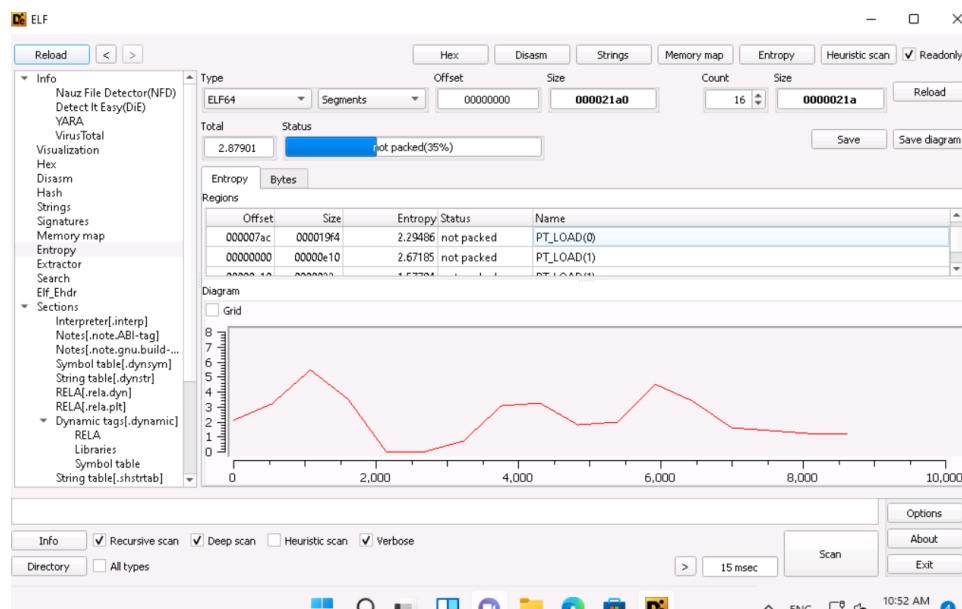


Figure 10 — Section entropy analysis showing the binary is not packed (overall packer confidence 35%).

- Entropy across loaded segments stayed within a range consistent with unpacked, non-obfuscated code, supporting straightforward static analysis without needing to first unpack the sample.

5. Network Behavior & Detection

3.11 Live Connection Monitoring (TCPView)

Process Name	Process ID	Protocol	State	Local Address	Local Port	Remote Address	Remote Port	Create Time
dns.exe	3564	TCP	Listen	10.10.1.22	53	0.0.0.0	0	8/2/2026 11:18:00
dns.exe	3564	TCP	Listen	127.0.0.1	53	0.0.0.0	0	8/2/2026 11:18:00
svchost.exe	520	TCP	Listen	0.0.0.0	135	0.0.0.0	0	8/2/2026 11:17:47
System	4	TCP	Listen	10.10.1.22	139	0.0.0.0	0	8/2/2026 11:17:46
lsass.exe	820	TCP	Listen	0.0.0.0	389	0.0.0.0	0	8/2/2026 11:17:59
svchost.exe	520	TCP	Listen	0.0.0.0	593	0.0.0.0	0	8/2/2026 11:17:59
lsass.exe	820	TCP	Listen	0.0.0.0	606	0.0.0.0	0	8/2/2026 11:17:59
mqsvc.exe	3760	TCP	Listen	0.0.0.0	1801	0.0.0.0	0	8/2/2026 11:18:01
mqsvc.exe	3760	TCP	Listen	0.0.0.0	2103	0.0.0.0	0	8/2/2026 11:18:01
mqsvc.exe	3760	TCP	Listen	0.0.0.0	2105	0.0.0.0	0	8/2/2026 11:18:01
mqsvc.exe	3760	TCP	Listen	0.0.0.0	2107	0.0.0.0	0	8/2/2026 11:18:01
lsass.exe	820	TCP	Listen	0.0.0.0	3268	0.0.0.0	0	8/2/2026 11:18:29
lsass.exe	820	TCP	Listen	0.0.0.0	3269	0.0.0.0	0	8/2/2026 11:18:29
svchost.exe	1052	TCP	Listen	0.0.0.0	3389	0.0.0.0	0	8/2/2026 11:17:50
postgres.exe	4628	TCP	Listen	127.0.0.1	7337	0.0.0.0	0	8/2/2026 11:18:18
Microsoft.ActiveDirec...	3612	TCP	Listen	0.0.0.0	9389	0.0.0.0	0	8/2/2026 11:18:30
lsass.exe	820	TCP	Listen	0.0.0.0	49664	0.0.0.0	0	8/2/2026 11:17:47
wininit.exe	632	TCP	Listen	0.0.0.0	49665	0.0.0.0	0	8/2/2026 11:17:47
svchost.exe	1340	TCP	Listen	0.0.0.0	49666	0.0.0.0	0	8/2/2026 11:17:47
svchost.exe	2012	TCP	Listen	0.0.0.0	49667	0.0.0.0	0	8/2/2026 11:17:50
svchost.exe	2588	TCP	Listen	0.0.0.0	49668	0.0.0.0	0	8/2/2026 11:17:50
lsass.exe	820	TCP	Listen	0.0.0.0	49669	0.0.0.0	0	8/2/2026 11:17:50
svchost.exe	2872	TCP	Listen	0.0.0.0	57108	0.0.0.0	0	8/2/2026 11:17:51
lsass.exe	820	TCP	Listen	0.0.0.0	60650	0.0.0.0	0	8/2/2026 11:17:59
spoolsv.exe	3448	TCP	Listen	0.0.0.0	60651	0.0.0.0	0	8/2/2026 11:18:00
mqsvc.exe	3760	TCP	Listen	0.0.0.0	60654	0.0.0.0	0	8/2/2026 11:18:01
services.exe	776	TCP	Listen	0.0.0.0	60657	0.0.0.0	0	8/2/2026 11:18:03
dns.exe	3564	TCP	Listen	0.0.0.0	60673	0.0.0.0	0	8/2/2026 11:18:30
dfs.exe	3572	TCP	Listen	0.0.0.0	60695	0.0.0.0	0	8/2/2026 11:18:31
System	4	TCP	Established	10.10.1.22	60718	10.10.1.11	445	8/2/2026 11:18:51

Figure 11 — TCPView listing active TCP listeners and connections on the lab target.

- Standard host services (DNS, LDAP/lsass, RDP on 3389, etc.) are visible for context alongside an established TCP session from the lab target (10.10.1.22) back to the attacker host (10.10.1.11), consistent with an active RAT callback.
- This demonstrates that even a lightweight, minimally-obfuscated RAT stub produces an observable network fingerprint that endpoint/network monitoring can detect — reinforcing the value of egress filtering and connection auditing on production networks.

6. Key Findings

- The RAT builder produces a small (~3 KB), unpacked PE stub, which is nonetheless detected by the large majority of AV engines and sandboxes due to well-established signature and behavioral coverage of this tool family.
- Post-infection capability observed in the exercise included process enumeration/control, file system browsing, registry access, and direct operator-to-victim messaging — a broad post-compromise toolkit for a single payload.
- The tool maintains a persistent outbound TCP connection to its configured callback host/port, which is clearly visible with standard connection-monitoring tools (TCPView) and would be catchable by network egress monitoring or EDR in a production environment.
- The exercise reinforced that even unsophisticated commodity RATs pose meaningful risk when antivirus/EDR is absent, disabled, or the binary is delivered via user-driven social engineering.

7. Defensive Recommendations

- Maintain up-to-date antivirus/EDR on all endpoints; the sample tested here was caught by the majority of engines, underscoring the value of baseline AV coverage.
- Deploy egress filtering / network monitoring to flag unexpected outbound connections to unfamiliar hosts and non-standard ports.
- Enforce application allow-listing and disable execution from user-writable temp directories where feasible, since builder defaults frequently target %TEMP%.
- Deliver regular user security-awareness training, since the initial infection vector for commodity RATs is typically phishing or social engineering rather than a technical exploit.
- Monitor for anomalous registry and process-management API activity as a behavioral detection layer, complementing signature-based AV.

8. Scope & Ethical Use Statement

All activity described in this report was carried out within an isolated, dedicated training lab environment (private 10.10.1.0/24 range, CEH.com training domain) with no production systems, real user data, or external networks involved. The payment-card number referenced in Figure 6 was a fabricated placeholder used only to illustrate a data-exposure scenario. This material is intended solely for authorized security-education purposes and should not be replicated outside a controlled, consent-based lab environment.

End of Report